1.2.1 Rational numbers on a number line

Locating the rational numbers on a number line is an important skill. For example, to represent the number $\frac{-3}{4}$ on the number line, $\frac{-3}{4}$ being negative would be marked to the left of 0 and it is between 0 and –1. We know that the integers, 1 and –1 are equidistant from 0 and so are the numbers 2 and –2, 3 and –3 from 0. This concept remains the same for rational numbers too. Now, as we mark $\frac{3}{4}$ to the right of zero, at 3 parts out of 4 between 0 and 1, the same way, we will mark $\frac{-3}{4}$ to the left of zero, at 3 parts out of 4 between 0 and –1 as shown below.

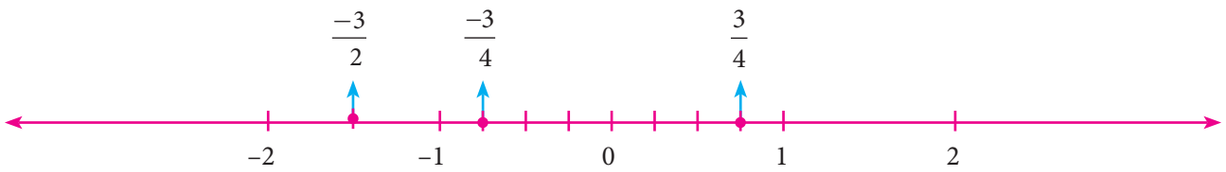


Fig. 1.5

Similarly, it is easy to find $\frac{-3}{2}$ between –1 and –2 since $\frac{-3}{2} = -1\frac{1}{2}$.

Now, on the following number line what rational numbers do the letters A and B represent?



Fig. 1.6

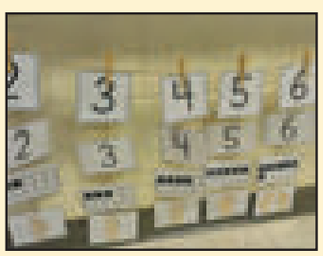
You will now be able to say easily the rational numbers marked by A and B on the number line as shown above. Isn't it? Here, A represents the rational number $-4\frac{4}{7}\left(\frac{-32}{7}\right)$ and B represents the rational number $3\frac{3}{5}\left(\frac{18}{5}\right)$.

1.2.2 Decimal representation of a rational number

A rational number can be nicely represented in decimal form rather than in the usual fractional form. Given a rational number in the form $\frac{a}{b}$ ($b \neq 0$), just divide the numerator a by the denominator b and we can see that it can be expressed as a terminating or non-terminating, recurring decimal.

Activity

Use a string as a number line and fix it on the wall, for the length of the class room. Just mark the integers spaciously and ask the students to pick the rational number cards from a box and fix it roughly at the right place on the number line string. This can be played between teams and the team which fixes more number of cards correctly (by marking) will be the winner.



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